

DIPANSHU HALDAR

Senior Data Scientist — Applied ML (Graph Learning, LLMs, Causal Inference)

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Summary

Senior Data Scientist with 11+ years building applied ML systems that ship to production, across banking, adtech, and gaming (Electronic Arts). Current focus: graph representation learning (GNNs, PyTorch Geometric), transformer-based NLP, and causal inference applied to player behavior, trust & safety, and growth. Comfortable owning a problem end-to-end — from causal design through production deployment on AWS.

Experience

Electronic Arts Inc.

Senior Data Scientist

May 2020 – Present

Hyderabad, India

- **Graph ML for player networks:** Built graph neural network models (PyTorch Geometric, graph embeddings, Node2Vec + clustering) for community and influencer detection in Madden's multiplayer network, surfacing high-impact player clusters to inform engagement strategy.
- **Link prediction for re-engagement:** Combined graph embeddings with panel econometrics to estimate player re-engagement probability, achieving AUC 0.70–0.74 on multiplayer match data.
- **Transformer-based trust & safety:** Applied Hugging Face transformer architectures (BERT) to build toxicity detection models used in production.
- **Causal inference for growth:** Led a marketing metric-maturity initiative combining Difference-in-Differences, Causal Impact, ridge/elastic-net regression, and Marketing Mix Modeling to separate true campaign lift (~1–4%) from noise; findings embedded directly into EA's growth strategy and spend allocation.
- **Forecasting at scale:** Owned demand forecasting for Customer Experience contact volume; evaluated Prophet, TBATS, and Temporal Fusion Transformer, reducing forecast error from ~30% to ~25% MAPE; shipped as a self-serve Shiny dashboard used org-wide.
- **Account security ML:** Built and deployed a Random Forest account-takeover detection system on AWS SageMaker (S3, Lambda, endpoints), achieving ~75% precision / 65–70% recall in production for FIFA.

Datametica Solutions Pvt. Ltd.

Senior Data Scientist

Aug 2018 – Feb 2020

Pune, India

- Built an NLP classification pipeline to detect rumor vs. verified news for a stock exchange client; SVM model reached 76–80% accuracy (F1 ~0.74) using frequency-based text embeddings.

Development Bank of Singapore (DBS)

Lead Engineer, Data Science – Institutional Banking

Jul 2016 – Aug 2018

Hyderabad, India

- Built a word-embedding-based system (Word2Vec) to auto-match Letter of Credit and commercial invoice metadata, flagging discrepancies in buyer/seller and quantity information.
- Built anomaly detection for over/under-invoicing by forecasting expected traded-goods pricing and comparing against disbursed amounts.
- Built credit-risk scoring pipeline (Apache Spark, R) parsing bank statements and credit bureau reports (CIBIL, Moody's) to compute customer credit scores.
- Used Cosine and Levenshtein similarity on transaction data (SparkR) to detect fraudulent/biased supplier concentration across the APAC region.

Inside View Technologies (now Demandbase)

Data Analyst

Mar 2015 – Jul 2016

Hyderabad, India

- Built NLP systems for company news similarity (Word2Vec, cosine similarity) and Named Entity Recognition for product/executive-persona extraction from news and employee data.

HDFC Bank Ltd.

Deputy Manager, Business Intelligence – Digital Banking

Nov 2014 – Mar 2015

Chennai, India

- Built ARIMA-based ATM cash demand forecasting models for two Chennai regions, achieving ~25–28% MAPE.

Aaum Research and Analytics Pvt. Ltd. (IIT Madras)

Data Scientist

Mar 2014 – Nov 2014

Chennai, India

- Built marketing mix models and recommendation engines (collaborative filtering) in R, including a market-mix model explaining ~55–65% of sales variance and a dating-app recommender with ~58–63% Top-5 accuracy.

Technical Skills

Deep Learning & Graph ML: PyTorch, PyTorch Geometric, Graph Neural Networks, Node2Vec, Transformers (Hugging Face, BERT), ANN, RNN, LSTM

Causal Inference & Econometrics: Difference-in-Differences, RDD, Instrumental Variables, Matching, Causal Impact, Marketing Mix Modeling

Machine Learning: Random Forest, XGBoost, SVM, Regression (Linear, Logistic, GAM), Clustering (DBSCAN, Louvain)

Time Series: ARIMA, Prophet, TBATS, BSTS, N-HiTS, Temporal Fusion Transformer

Languages & Tools: Python, R, SQL, Shell, scikit-learn, TensorFlow, NetworkX, Tidyverse, Shiny

MLOps & Infra: AWS SageMaker, S3, Lambda, Apache Spark, Hive, Qubole, POSIT Workbench/Cloud

Education

University of Calcutta — M.Sc. Statistics

2007–2009

Presidency College, Kolkata — B.Sc. Statistics

2003–2006